Algebra 1 Name____________________
Unit 1 Date____________________
Lesson 8 Practice Period____________________

1. Which of the following is not a radical expression that is equivalent to $\sqrt{96}$?

A. $\sqrt{2}\cdot\sqrt{48}$
B. $\sqrt{5}\cdot\sqrt{19}$
C. $\sqrt{6}\cdot\sqrt{16}$
D. $\sqrt{8}\cdot\sqrt{12}$

$48 \times 2 = 96$ ✓ $19 \times 5 = 95$ ✗ $16 \times 6 = 96$ ✓ $12 \times 8 = 96$ ✓

2. Which of the expressions from #1 can be rewritten in the form of an integer multiplied by the radical? *Explain.*

Option C because 16 is a perfect square so $\sqrt{6}\cdot\sqrt{16} = 4\sqrt{6}$.

For questions 3-4, write two equivalent expressions for each radical expression using the product law of radicals. One of your expressions should simplify to an integer multiplied by a radical.

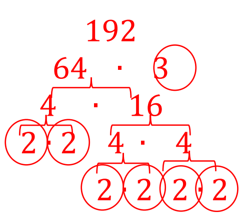
3. $\sqrt[3]{192} = \sqrt[3]{64}\cdot\sqrt[3]{3} = 4\sqrt[3]{3}$

$\sqrt[3]{192} = \sqrt[3]{96}\cdot\sqrt[3]{2}$

4. $\sqrt{80} = \sqrt{16}\cdot\sqrt{5} = 4\sqrt{5}$

$\sqrt{80} = \sqrt{2}\cdot\sqrt{40}$

5. Rewrite $\sqrt[3]{192}$ as an integer multiplied by a radical.

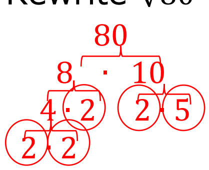


$\sqrt[3]{2\cdot2\cdot2\cdot2\cdot2\cdot2\cdot3}$

$2\cdot2\sqrt[3]{3}$

$= 4\sqrt[3]{3}$

6. Rewrite $\sqrt{80}$ as an integer multiplied by a radical.



$\sqrt{2\cdot2\cdot2\cdot2\cdot5}$

$2\cdot2\sqrt{5}$

$= 4\sqrt{5}$

7. Identify all expressions that are equivalent to $2\sqrt[3]{54}$.

- $6\sqrt[3]{2}$
- $23\sqrt[3]{2}$
- $6\sqrt[3]{6}$
- $2 \cdot \sqrt[3]{2} \cdot \sqrt[3]{27}$
- $2 \cdot \sqrt[3]{6} \cdot \sqrt[3]{9}$

54
3 18
3 · 6
2 · 3

$2 \cdot \sqrt[3]{2 \cdot 3 \cdot 3 \cdot 3}$
$2 \cdot 3 \cdot \sqrt[3]{2}$
$= 6\sqrt[3]{2}$

Factors of 54: $1 \cdot 54$
$2 \cdot 27$
$3 \cdot 18$
$6 \cdot 9$

For questions 8-10, write an equivalent expression in the form of a constant multiplied by a radical.

8. $2\sqrt{200}$

$2\sqrt{100 \cdot 2}$
$10 \cdot 2\sqrt{2}$
$= 20\sqrt{2}$

9. $\sqrt[3]{40}$

40
4 · 10
2 2 2 · 5

$\sqrt[3]{2 \cdot 2 \cdot 2 \cdot 5}$

$= 2\sqrt[3]{5}$

10. $8\sqrt[3]{(-500)}$

$8\sqrt[3]{(-125) \cdot 4}$

$-5 \cdot 8\sqrt[3]{4}$

$= -40\sqrt[3]{4}$